

The code for \$a_2\$ is:

Please provide a brief explanation of your answer.

No, it is not true.

The code defines a_2 as the limit of $\text{Compute } a_{2(n)} \text{ as } n \rightarrow \infty$.

Evaluating the sum in `Compute_f(n)` :

- The initial term is $72^{-10^{24}}$, which is extremely small.
- The loop adds $\sum_{k=3}^{n+2} \left((1/2)^k - 72^{-10^{23} \cdot k} \right)$.

As $n \rightarrow \infty$, the sum of $(1/2)^k$ is $1/4$, and the sum of $72^{-10^{23} \cdot k}$ is approximately $72^{-3 \cdot 10^{23}}$, which is much larger than the initial $72^{-10^{24}}$ term.

Therefore, $f(\infty) \approx \frac{1}{4} - 72^{-3 \cdot 10^{23}}$ (slightly less than $1/4$), and so $a_2 = 2f(\infty) + \frac{1}{2} \approx 1 - \varepsilon$ with $\varepsilon > 0$. Hence $a_2 < 1$.

Minimising total treatment time $T = x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$ and $x_1, x_2 \geq 0$:

Since $a_2 < 1$, the objective is minimised by making x_2 as small as possible, i.e., $x_2 = 0$, $x_1 = 1$. The unique optimal minimiser is $(1, 0)$.

The infinity-norm distance from Jordan's point $(0.3, 0.7)$ to $(1, 0)$ is $\max(|0.3 - 1|, |0.7 - 0|) = 0.7$, which is **not** ≤ 0.4 .

(Note: The revelation about $(0.6, 0.4)$ being within 0.45 of the minimiser is true, as its distance is 0.4.)